

Developer Mock Test Results

Test ID:	a13703ec-d0d2-48...	Student ID:	64644
Test Type:	Developer Assessment	Date:	2026-03-04 13:13:05
Overall Score:	11/25 (44.0%)	Performance:	■ Average
Warnings:	0/3	Status:	Completed ■

Section-wise Performance

Section	Score	Percentage	Status
■ Aptitude	2/10	20.0%	■■ Needs Work
■ MCQ/Theory	5/10	50.0%	■ Pass
■ Coding	4/5	80.0%	■ Pass

Detailed Question Review

■ APTITUDE SECTION (2/10 - 20.0%)

■ Q1. If a number is decreased by 20% and then increased by 20%, what is the overall percentage change?

Your Answer: 0%

Correct Answer: -4%

■ The correct answer is -4% because when a number is decreased by 20% and then increased by 20%, the overall change is not 0%. To see this, let's calculate: if we start with 100, decreasing by 20% gives 80, and then increasing 80 by 20% gives 96, which is 4% less than the original 100. The user's mistake was likely assuming that the two 20% changes would cancel each other out, which is not the case due to the different bases for the percentage changes.

■ Q2. What is the next number: 1, 3, 6, 10, 15, ?

Your Answer: 20

Correct Answer: 21

■ The correct answer is 21 because the pattern of the sequence is formed by adding 2, 3, 4, 5, and so on to the previous term. The key calculation step is: $1 + 2 = 3$, $3 + 3 = 6$, $6 + 4 = 10$, $10 + 5 = 15$, $15 + 6 = 21$. The user's likely mistake was not recognizing the incremental increase in the differences between consecutive terms.

■ Q3. The average of 2 numbers is 18. What is the sum of the 2 numbers?

Your Answer: 36

■ Correct! Well done.

■ Q4. A shopkeeper buys an item at Rs.200 and sells it at a 10% loss. What is the selling price?

Your Answer: Rs.170

Correct Answer: Rs.190

■ The correct answer is Rs.180, which can be calculated as follows: $\text{Selling Price} = \text{Cost Price} - \text{Loss} = \text{Rs.200} - (10\% \text{ of Rs.200}) = \text{Rs.200} - \text{Rs.20} = \text{Rs.180}$. The user's answer, Rs.170, is likely incorrect because they may have miscalculated the loss amount or subtracted a larger amount from the cost price. The correct calculation shows that the selling price is Rs.180, not Rs.190 or any other option.

■ **Q5. A shopkeeper buys 100 items at Rs.10 each and sells them at Rs.12 each. What is the profit percentage?**

Your Answer: 20%

■ Excellent! Right answer.

■ **Q6. If 3 men can build a wall in 8 days, how many men are needed to build the same wall in 4 days?**

Your Answer: 8

Correct Answer: 6

■ To determine the number of men needed to build the wall in 4 days, we can use the formula: $\text{men} * \text{days} = \text{constant}$. Given that 3 men can build the wall in 8 days, we have $3 * 8 = 24$ man-days. To build the same wall in 4 days, we need to find the number of men that satisfies the equation: $\text{men} * 4 = 24$. Solving for men, we get $\text{men} = 24 / 4 = 6$. The user's likely mistake was not using this calculation step to find the correct number of men.

■ **Q7. If it is true that all cars are vehicles and some vehicles are bicycles, what can be concluded?**

Your Answer: All bicycles are cars

Correct Answer: Some vehicles are cars

■ The correct answer is "Some vehicles are cars" because we know that all cars are vehicles, which implies that cars are a subset of vehicles. The user's answer, "All bicycles are cars", is incorrect because the original statement only mentions that some vehicles are bicycles, but it does not establish a relationship between bicycles and cars. The user likely made a mistake by incorrectly assuming a direct relationship between bicycles and cars, rather than recognizing the subset relationship between cars and vehicles.

■ **Q8. If it is true that all dogs are animals and some animals are cats, what can be concluded?**

Your Answer: No dogs are cats

Correct Answer: Some animals are dogs

■ The correct answer is "Some animals are dogs" because we know that all dogs are animals, which directly implies that some animals (at least the ones that are dogs) are indeed dogs. The user's answer, "No dogs are cats", is an incorrect conclusion because the given statements do not provide enough information to make this claim. The user's mistake likely stems from incorrectly assuming that the categories of dogs and cats are mutually exclusive based on the given information.

■ **Q9. If it is true that all books are written and some written things are articles, what can be concluded?**

Your Answer: No books are articles

Correct Answer: Some written things are books

■ The correct answer is "Some written things are books" because we know that all books are written, which means books are a subset of written things. The user's answer, "No books are articles", is incorrect because the information provided does not rule out the possibility of some books being articles. The user likely made a mistake by assuming that "some written things are articles" implies that books cannot be articles, when in fact it only implies that there are other types of written things besides books.

■ **Q10. A mixture is made of 2 parts oil and 5 parts water. What is the ratio of oil to the total mixture?**

Your Answer: 1/2

Correct Answer: 2/7

■ The correct answer is 2/7 because the total mixture is made of 2 parts oil and 5 parts water, making a total of $2 + 5 = 7$ parts. The ratio of oil to the total mixture is then 2/7. The user's likely mistake was assuming the ratio was a simple half, but the calculation $2/(2+5) = 2/7$ shows that oil makes up only 2 out of 7 parts of the total mixture.

■ MCQ SECTION (5/10 - 50.0%)

■ **Q1. What type of programming language is Java?**

Your Answer: High-level, object-oriented

■ *Correct! Good understanding.*

■ **Q2. What is the next step after creating a new Java file?**

Your Answer: Open a code editor or IDE

Correct Answer: Write your code

■ *The correct answer is "Write your code" because after creating a new Java file, the next logical step is to start writing the code that will be executed by the Java program. The user's answer, "Open a code editor or IDE", is likely incorrect because this step would have already been completed in order to create the new Java file. The user probably got it wrong because they thought about the step that comes before creating a new Java file, rather than the step that comes after.*

■ **Q3. What type of error occurs when a program has a syntax mistake?**

Your Answer:

Runtime exception

Correct Answer:

Syntax error

■ *The correct answer is A) Syntax error because it occurs when a program has a syntax mistake, meaning the code does not follow the language's rules and structure. The user's answer, Runtime exception, is incorrect because it refers to an error that occurs during the execution of the program, not during the compilation or interpretation phase when syntax is checked. The user likely mistook the two concepts, as runtime exceptions happen when the code is running, whereas syntax errors are caught before the code is executed.*

■ **Q4. What is a good practice for naming classes in Java?**

Your Answer: Use meaningful and descriptive names

■ *Well done!*

■ **Q5. What is a good practice for keeping code organized in Java?**

Your Answer: Using methods and variables

Correct Answer: Using classes and packages

■ *The correct answer, "Using classes and packages", is the right choice because it allows for a high level of organization and structure in Java code, enabling developers to group related classes and functionality together. This promotes modularity, reusability, and maintainability of the code. The user's answer, "Using methods and variables", is incorrect because while methods and variables are essential components of Java programming, they are not sufficient on their own to keep code organized at a higher level, which is where classes and packages come into play.*

■ **Q6. What is the purpose of a method that takes two numbers and returns their sum?**

Your Answer: To perform a specific task

■ *Well done!*

■ **Q7. What type of block is used to handle errors in Java?**

Your Answer: Try-catch block

■ *Correct! Good understanding.*

■ **Q8. What are the types of control structures in Java?**

Your Answer: Methods and classes

Correct Answer: Conditions and loops

■ *The correct answer, "Conditions and loops", is right because control structures in Java refer to the ways in which the flow of a program's execution is managed, and conditions (such as if-else statements) and loops (such as for and while loops) are the primary means of doing so. The user's answer, "Methods and classes", is incorrect because while methods and classes are fundamental components of Java programming, they are not types of control structures. The user likely mistook these essential programming concepts for control structures, which are specifically used to direct the program's flow.*

■ **Q9. What type of programming paradigm does Java follow?**

Your Answer: Procedural programming

Correct Answer: Object-oriented programming

■ The correct answer is object-oriented programming because Java is designed around the principles of encapsulation, inheritance, and polymorphism, which are fundamental to object-oriented programming. The user's answer, procedural programming, is likely incorrect because while Java can be used in a procedural style, its core design and syntax are centered around objects and classes. The user may have mistakenly associated Java with procedural programming due to its ability to support procedural elements, but its primary paradigm is object-oriented.

■ Q10. What are the main parts of a Java program?

Your Answer: Methods, classes, objects

■ Well done!

■ CODING SECTION (4/5 - 80.0%)

■ Q1. Write a program that reads two integers and prints their sum, difference, product, and quotient.

****Input Format:**** - Line 1: First integer (integer) ...

Your Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        solution();
    }

    public static void solution() {
        Scanner scanner = new Scanner(System.in);
        int a = Integer.parseInt(scanner.nextLine().trim());
        int b = Integer.parseInt(scanner.nextLine().trim());

        int sum      = a + b;
        int difference = a - b;
        int product   = a * b;
        double quotient = (double) a / b;

        String quotientStr = (quotient == Math.floor(quotient))
            ? String.format("%.1f", quotient)
            : String.valueOf(quotient);

        System.out.println(sum + " " + difference + " " + product + " " + quotientStr);
    }
}
```

■ Your code has successfully passed all test cases, indicating that it is correct and functions as expected. Great job on writing robust and accurate code that meets the requirements.

■ Q2. Write a program that reads a string and prints the string with all characters converted to uppercase, except for the first character which should be i...

Your Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        solution();
    }

    public static void solution() {
        Scanner scanner = new Scanner(System.in);
        String input = scanner.nextLine().trim();
    }
}
```

```

    if (input.isEmpty()) {
        System.out.println("");
        return;
    }

    String[] words = input.split("\\s+");

    StringBuilder result = new StringBuilder();

    for (int i = 0; i < words.length; i++) {
        if (i == 0) {
            // First word → all lowercase
            result.append(words[i].toLowerCase());
        } else {
            // Rest of words → all uppercase
            result.append(words[i].toUpperCase());
        }
        if (i < words.length - 1) result.append(" ");
    }

    System.out.println(result.toString());
}

```

Correct Answer:

```

s = input()
if len(s) > 0:
    print(s[0].lower() + s[1:].upper())
else:
    print(s)

```

■ There is no test case failure provided to analyze. Please provide the test case results to determine what went wrong with the student's code. Without the test case results, it's impossible to identify the issue. Please share the test case output to provide accurate feedback.

■ **Q3. Write a program that simulates a shopping cart, allowing users to add items, remove items, and calculate the total cost. ****Input Format:**** - Line 1: ...**

Your Answer:

```

import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        solution();
    }

    public static void solution() {
        Scanner scanner = new Scanner(System.in);

        int n = Integer.parseInt(scanner.nextLine().trim());
        double total = 0;

        for (int i = 0; i < n; i++) {
            String[] parts = scanner.nextLine().trim().split("\\s+");
            double price = Double.parseDouble(parts[1]);
            total += price;
        }

        if (total == Math.floor(total)) {
            System.out.println((int) total);
        } else {

```

```

        System.out.println(total);
    }
}
}

```

■ Your code has successfully passed all test cases, indicating that it is correct and functions as expected. Great job on writing robust and accurate code!

■ **Q4. Write a program that reads a dictionary of word frequencies and prints the word with the highest frequency. **Input Format:** - Line 1: Number of wor...**

Your Answer:

```

import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        solution();
    }

    public static void solution() {
        Scanner scanner = new Scanner(System.in);

        int n = Integer.parseInt(scanner.nextLine().trim());
        String topWord = "";
        int topFreq = -1;

        for (int i = 0; i < n; i++) {
            String[] parts = scanner.nextLine().trim().split("\\s+");
            String word = parts[0];
            int freq = Integer.parseInt(parts[1]);

            if (freq > topFreq) {
                topFreq = freq;
                topWord = word;
            }
        }

        System.out.println(topWord);
    }
}

```

■ Your code has successfully passed all test cases, indicating that it is correct and functions as intended. Great job on writing robust and accurate code!

■ **Q5. Write a program that reads a string and prints the string with all characters converted to lowercase. **Input Format:** - Line 1: Input string (strin...**

Your Answer:

```

import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        solution();
    }

    public static void solution() {
        Scanner scanner = new Scanner(System.in);
        String input = scanner.nextLine();
        System.out.println(input.toLowerCase());
    }
}

```

■ Your code has successfully passed all test cases, indicating that it is correct and functions as expected. Great job on writing robust and accurate code!

■ Recommendations

Areas that need improvement:

- **Aptitude:** Practice more logical reasoning, quantitative aptitude, and problem-solving questions.